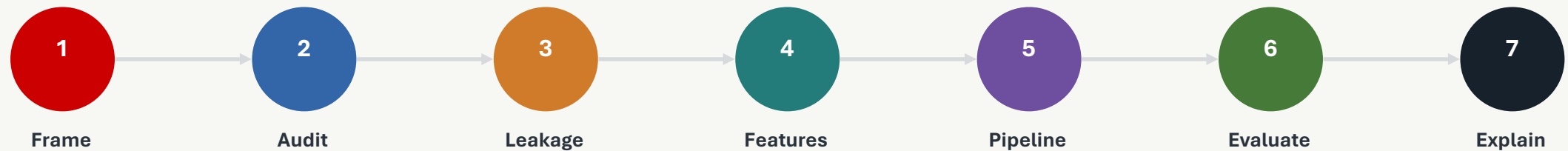


DAY 5

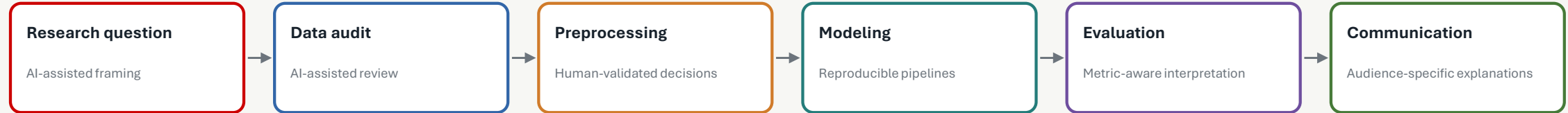
AI Across the Full Machine-Learning Pipeline

framing, auditing, leakage, features, pipelines, evaluation, and communication



Why use AI in the ML pipeline?

AI helps structure, accelerate, and communicate the workflow—but does not replace validation.



AI is useful when it...

- asks clarifying questions about the problem
- turns analysis outputs into interpretable summaries
- checks for missing logic, leakage, and metric mismatch
- helps draft code, documentation, and limitations

AI is risky when it...

- invents dataset characteristics or model results
- treats correlation as causation
- overrides domain knowledge or required inspection
- hides uncertainty behind fluent explanations

Day 5 focus: AI as a copilot for defensible analysis and evidence-grounded communication.

Problem Framing

Use AI to turn a broad research idea into a precise ML task.

What AI can help define

- unit of observation
- candidate inputs and target
- task type and output format
- baseline and primary metric

Human checks

- is the target measurable?
- are inputs available at prediction time?
- what decision changes?
- what error is most costly?

Goal: ask the AI to identify assumptions, then verify or correct each one.

Topic 1 prompt: frame the ML problem

Use prompt templates to turn AI into a structured reviewer—not an unchecked answer generator.

Prompt template

I have process temperature, pressure, speed, vibration, material type, and a defect label.

Help me define:

1. unit of observation
2. target variable
3. candidate inputs
4. ML task type
5. simple baseline
6. primary evaluation metric

Ask me to verify assumptions, especially which variables are available before prediction.

- AI should not choose the final problem
- the user must verify timing and target meaning
- baseline and metric must match the decision

Data Audit

AI can review a DataFrame summary and suggest issues to investigate.

Supply the AI with

- `df.shape` and column names
- `df.dtypes`
- `df.isna().sum()`
- duplicates and target balance
- summary statistics and category counts

Ask AI to identify

- missing values and invalid types
- class imbalance
- possible identifiers
- suspicious ranges or categories
- questions for a domain expert

Avoid: “clean my data.” Prefer: “review this audit and list concerns without deleting observations unless a domain rule supports it.”

Topic 2 prompt: audit the dataset

Use prompt templates to turn AI into a structured reviewer—not an unchecked answer generator.

Prompt template

Review this data audit. Identify potential data-quality concerns, but do not recommend deleting observations without a domain-based reason.

Dataset summary:

- shape: [paste df.shape]
- dtypes: [paste df.dtypes]
- missing values: [paste df.isna().sum()]
- duplicates: [paste duplicate count]
- target distribution: [paste class counts]
- numeric summary: [paste df.describe()]

- separate facts from recommendations
- flag issues, then verify in code
- ask whether unusual values are valid rare events

Leakage Review

AI can help ask the temporal question: was this variable available at prediction time?

Common leakage variables

- final inspection score
- maintenance action after run
- future sensor readings
- target-derived status fields
- group IDs that reveal labels

Evaluation Questions

- when is the prediction made?
- when is each feature known?
- is the test set too familiar?
- does a variable reveal the outcome?
- what would the real workflow see?

Leakage detection is one of the highest-value AI audit tasks because fluent models often miss temporal logic unless prompted explicitly.

Topic 3 prompt: check for leakage

Use prompt templates to turn AI into a structured reviewer—not an unchecked answer generator.

Prompt template

Here are the dataset columns and definitions. The prediction must be made before final inspection.

[paste column list and definitions]

Which variables could introduce target leakage? For each variable, explain:

- whether it is available before prediction
- how it could reveal the target
- whether to keep, transform, or remove it

Do not assume timing; list questions that need domain verification.

- final decisions require domain workflow knowledge
- leakage can make performance meaningless

Feature Engineering

Use AI to generate feature hypotheses, then validate them with domain knowledge.

AI-generated candidates

- temperature deviation from nominal
- pressure deviation from target
- vibration-to-speed ratio
- process intensity
- rolling statistics for time-series data

Human validation

- physical plausibility
- availability before prediction
- measurement reliability
- risk of redundancy or leakage
- interpretability for stakeholders

Key message: AI-generated features are hypotheses, not evidence that the feature improves the model.

Topic 4 prompt: propose features responsibly

Use prompt templates to turn AI into a structured reviewer—not an unchecked answer generator.

Prompt template

Suggest candidate features using temperature, pressure, speed, vibration, timestamp, and material type.

Separate features into:

1. domain-informed features
2. mathematically convenient features
3. risky or possibly leaky features

For each feature, state what assumption must be verified before using it. Do not claim that any feature will improve performance until tested.

- features should map to process logic
- test one change at a time
- report features that were tried and rejected

Pipeline Development

AI can draft code, but the analyst must inspect preprocessing order and data flow.

Useful AI coding support

- write scikit-learn pipelines
- explain code line-by-line
- debug shape/type errors
- refactor repeated steps
- add comments and documentation

Must inspect manually

- target and feature columns
- train/test split logic
- preprocessing fit only on training data
- metric functions and labels
- random seed and reproducibility

The code can run and still be scientifically wrong if the split, target, or feature timing is wrong.

Topic 5 prompt: create a reproducible pipeline

Use prompt templates to turn AI into a structured reviewer—not an unchecked answer generator.

Prompt template

```
Create a scikit-learn pipeline that:  
- uses a stratified train/test split  
- median-imputes and standardizes numerical variables  
- imputes and one-hot encodes categorical variables  
- fits logistic regression  
- compares against a majority-class baseline  
- reports accuracy, precision, recall, F1, and a confusion matrix
```

Use fixed random_state values. Explain each line briefly.

- verify preprocessing is inside the pipeline
- do not fit scalers on all data
- metric direction must match the target class

Evaluation and Error Analysis

AI can translate metrics into consequences when it is given the actual results and constraints.

Provide evidence

- baseline metrics
- model metrics
- confusion matrix
- class distribution
- primary error consequence
- known limitations

Ask for disciplined interpretation

- direct evidence
- reasonable interpretation
- unsupported claims
- most serious error
- recommended next analysis

AI should help answer: “What does this result support, and what should we avoid claiming?”

Topic 6 prompt: interpret results without overclaiming

Use prompt templates to turn AI into a structured reviewer—not an unchecked answer generator.

Prompt template

Interpret these results for a process engineer. Separate direct evidence, reasonable interpretation, and unsupported conclusions.

Baseline:

Accuracy = 0.80, Recall = 0.00

Model:

Accuracy = 0.86, Precision = 1.00, Recall = 0.27, F1 = 0.43

Confusion matrix:

TN = 44, FP = 0, FN = 8, TP = 3

Missed defects are more costly than false alarms.

- look for causal overclaims
- ensure recall is emphasized
- ask for next analysis, not deployment approval

Audience-Specific Communication

AI can translate the same evidence for researchers, managers, students, and reports.

Potential outputs

- technical interpretation
- manager summary
- student explanation
- model-card paragraph
- limitation statement
- next-step recommendation

Rules for communication

- use only supplied evidence
- state uncertainty clearly
- avoid causal language
- match audience and decision context
- separate model output from action

Generative AI is most useful after the analysis produces evidence; it organizes and explains, but should not create new evidence.

Topic 7 prompt: adapt the explanation

Use prompt templates to turn AI into a structured reviewer—not an unchecked answer generator.

Prompt template

Using only the supplied model results and limitations, create three versions of the interpretation:

1. technical explanation for a researcher
2. two-sentence summary for a lab manager
3. simple explanation for students new to ML

Each version must include the main limitation and avoid claiming causality.

- audience changes wording, not evidence
- AI can make unsupported claims sound polished
- require one “do not claim” statement

Takeaways: AI is an accelerator, not a substitute

The researcher remains responsible for the data, assumptions, evaluation, and conclusions.

AI accelerates

framing, code drafting,
summarization, and
documentation

AI needs evidence

metrics, tables, plots,
definitions, and limitations

AI must be audited

unsupported claims, leakage,
and causal language

Humans decide

validity, context, action, risk,
and ethics

Final message: AI can accelerate every stage of the ML workflow, but trustworthy use depends on human-validated evidence and clear boundaries.